Question 1

Non-idealities that increase the MOSFET I_{DS} when on vs the Shockley model.

The Shockley model is available in (1).

$$I_{DS} = \begin{cases} 0 & \text{if cutoff} \\ \beta\left(\left(V_{GS} - V_{TH}\right)V_{DS} - \frac{V_{DS}^2}{2}\right) & \text{if linear} \\ \frac{\beta}{2}\left(V_{GS} - V_{TH}\right)^2 & \text{if saturation} \end{cases} \tag{1}$$

$$I_{DS} = \frac{\beta}{2}\left(V_{GS} - V_{TH}\right)^2\left(1 + \lambda V_{DS}\right) \tag{2}$$

$$V_t = V_{t0} + \gamma\left(\sqrt{\phi_s + V_{sb}} - \sqrt{\phi_s}\right) \tag{3}$$

$$I_{DS} = I_{DS0}\exp\left(\frac{V_{GS} - V_{t0} + \eta V_{DS} - k_\gamma V_{sb}}{nv_T}\right)\left(1 - \exp\left(\frac{-V_{DS}}{v_T}\right)\right) \tag{4}$$

Channel length modulation Seen in (2), channel length modulation increases I_{DS} over the Shockley model.

Decreasing V_{TH} The different effects that decrease V_{TH} will increase $V_{GS} - V_{TH}$ which in turns increases the value of I_{DS}. There are some effects that decreases V_{TH}

Body effect By decreasing V_{sb}, especially below 0, (3) shall decrease V_{TH}, which then increases I_{DS}.

Short Channel effect By decreasing the channel length, V_{TH}.

Subthreshold leakage Technically, since the expected Shockley model at cutoff is $0\,\mathrm{A}$, then having a non-zero value is more. The subthreshold leakage is available in (4).

Question 2

The dual rail domino XOR-3 gate requires the complement function of a XOR-3. We are going to first define the definition for the regular XOR-3. The XOR-3 function outputs 1 when an odd number of inputs are 1, and 0 otherwise. The negation of the XOR-3 would be a XNOR-3, which outputs 1 when there is an even number of 1 in the input. This is described in (5), and (6).

$$\text{XOR-3} = A\overline{BC} + \overline{A}B\overline{C} + \overline{AB}C + ACB \tag{5}$$

$$\text{XNOR-3} = \overline{A}BC + A\overline{B}C + AB\overline{C} + \overline{ABC} \tag{6}$$

After finding these equations, the XOR-3 gate using dual rail domino logic is available in figure 3.

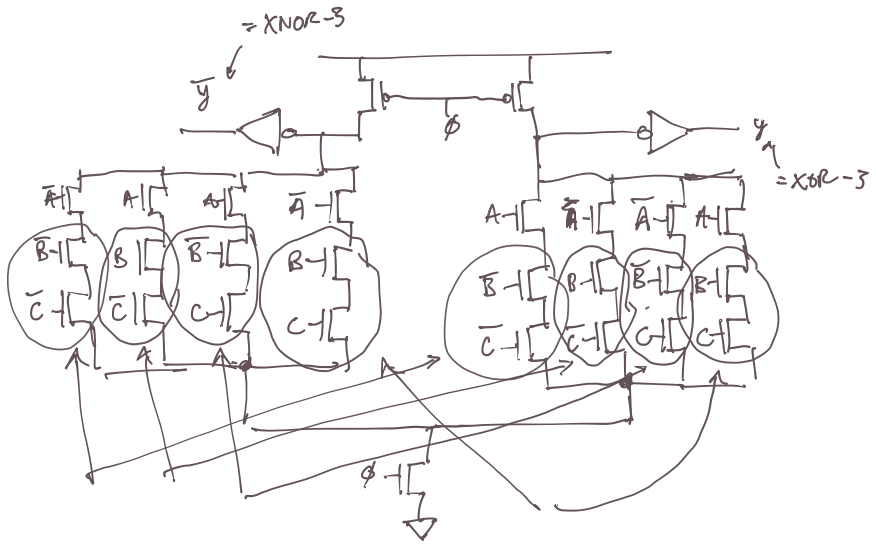

Figure 1: The unoptimized XOR-3 gate.

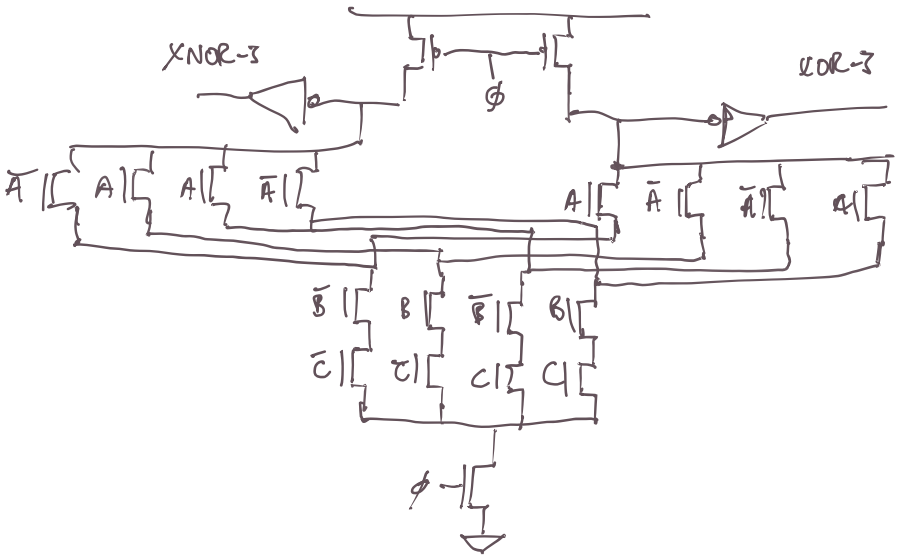

Figure 2: Intermediate XOR-3 gate.

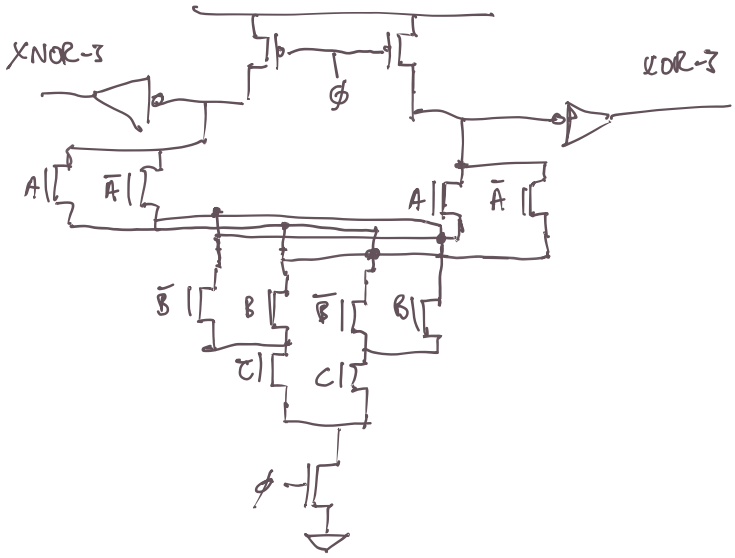

Figure 3: Optimized XOR-3 gate.